Questão 2

$$P(X=k) = \frac{e^{-\lambda} \lambda^{k}}{k!}$$

a) $P(X=6) = \frac{e^{-4} 4^{6}}{6!}$ $\therefore P(X=6) = \frac{e^{-4} \cdot 4096}{720}$

$P(X=6) = 0{,}01831 \cdot \frac{4096}{720} \cong 0{,}1042$ ou $10{,}42\%$

b) $P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

$\lambda = 4$ $\quad P(X=0) = \frac{e^{-4} 4^{0}}{0!} \cong 0{,}1831$

$P(X=1) = \frac{e^{-4} 4^{1}}{1!} = 4e^{-4} = 0{,}0732$

$P(X=2) = \frac{e^{-4} \cdot 4^{2}}{2!} = 8e^{-4} = 0{,}1465$

$P(X \leq 2) \cong 0{,}2381$ ou $23{,}81\%$

c) $\lambda = 4 \cdot 2 = 8$ $\quad Y \sim Poisson(8)$

$P(Y > 5) = 1 - P(Y \leq 5)$

$e^{-8} = 0,00033546$

$P(Y=0) = e^{-8} \cong 0,000335$

$P(Y=1) = 8e^{-8} \cong 0,00268$

$P(Y=2) = \frac{8^2}{2!} e^{-8} = \frac{64}{2} e^{-8} \cong 32e^{-8} \cong 0,0107$

$P(Y=3) = \frac{8^3}{3!} e^{-8} = \frac{512}{6} e^{-8} \cong 0,0286$

$P(Y=4) = \frac{8^4}{4!} e^{-8} = \frac{4096}{24} e^{-8} \cong 0,05725$

$P(Y=5) = \frac{8^5}{5!} e^{-8} = \frac{32768}{120} e^{-8} \cong 0,0916$

$P(Y \leq 5) \cong 0,1912$ $\quad P(Y \leq 5) = 1 - 0,1912$

$P(Y \leq 5) \cong 0,8088$ ou $80,88\%$

d) $E(x) = 4$ $\quad Var(x) = 4$ para 1 hor

$E(x) = 8$ $\quad Var(x) = 8$ para 2 horas